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# INTERGRATED THINKING IN COMPUTING



**PROJECT TITLE:**

**ECOSORT AI**

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# PROBLEM IDENTIFICATION

**Problem statement**

Kenya generates approximately 22,000 tones of wastes daily with Nairobi alone producing 2,400 tones per day. Only 10-15% of this waste is collected ,and even less is properly recycled, majority ends up in dumpsite like Dandora.

**POSSIBLE SOLUTIONS:**

- Improve waste collection systems especially in Nairobi and Mombasa which are mostly affected
- Develop an ECOSORT AI assistance that could help sort waste faster and more accurately.

# SELECTIVE INNOVATIVE SOLUTION

***DEV HUB*** **agreed to develop an AI assistance that can identify 20+ common Kenyan waste items with 85% accuracy including also;**

- ***Create a mobile app with camera scanning functionality.***
- ***Implement location-based rules.***
- ***Build a gamification system to encourage consistent recycling.***
- ***Design an analytic dashboard for tracking environmental impact.***

# Why ECOSORT AI

- Makes recycling easier and smart in Kenya.
- Helps sort waste right reducing landfill waste.
- Guides on what to recycle and reuse
- Support cleaner and greener communities.
- Encourages Eco-friendly habits with AI tips.

# Comparing Recycle Coach and Ecosort AI

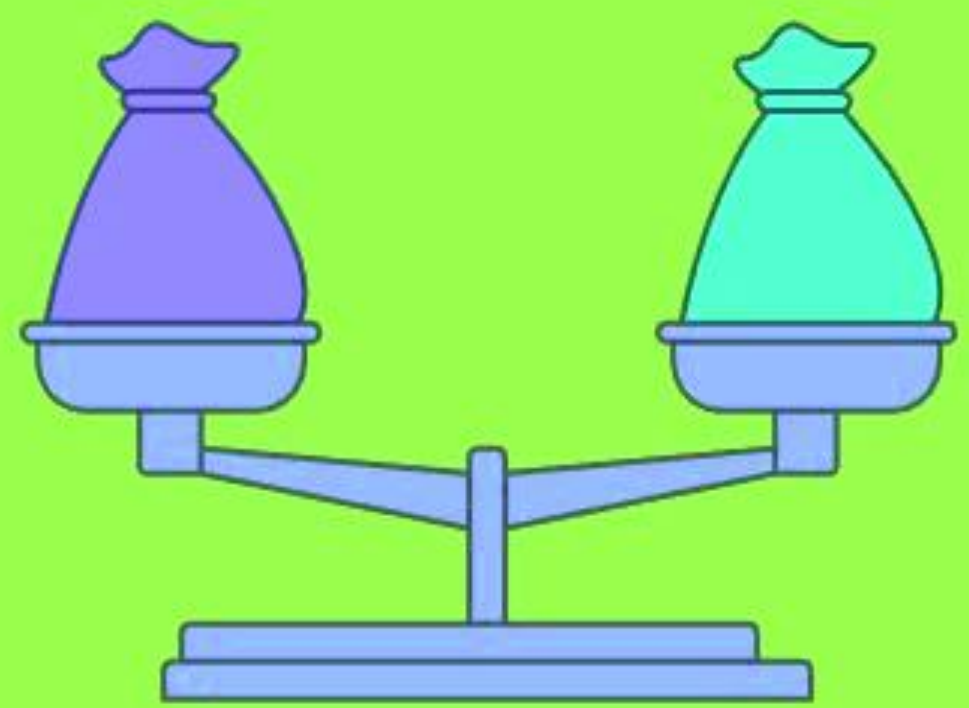
## Recycle Coach

### Focuses on garbage collection

Focuses on garbage collection

### Based abroad

Based abroad



## Ecosort AI

### Focuses on sorting of garbage

Focuses on sorting of garbage

### Based operations in Kenya

Based operations in Kenya

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# Sustainable Waste Management




- Medium to high investment
- Efficient waste reduction
- Adoption to growth
- AI-driven sorting and optimization

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# DATA SOURCE

**Where did the data come from?**

- Primary:Collected  within the university and Nchiru using smartphones.

**Type of data:**
Images of waste items + labels (plastic, paper, glass, metal, organic)

- **Brief description:**
  10+ images per category, captured in **real-world conditions** (different lighting, angles, backgrounds)

# DATA UNDERSTANDING

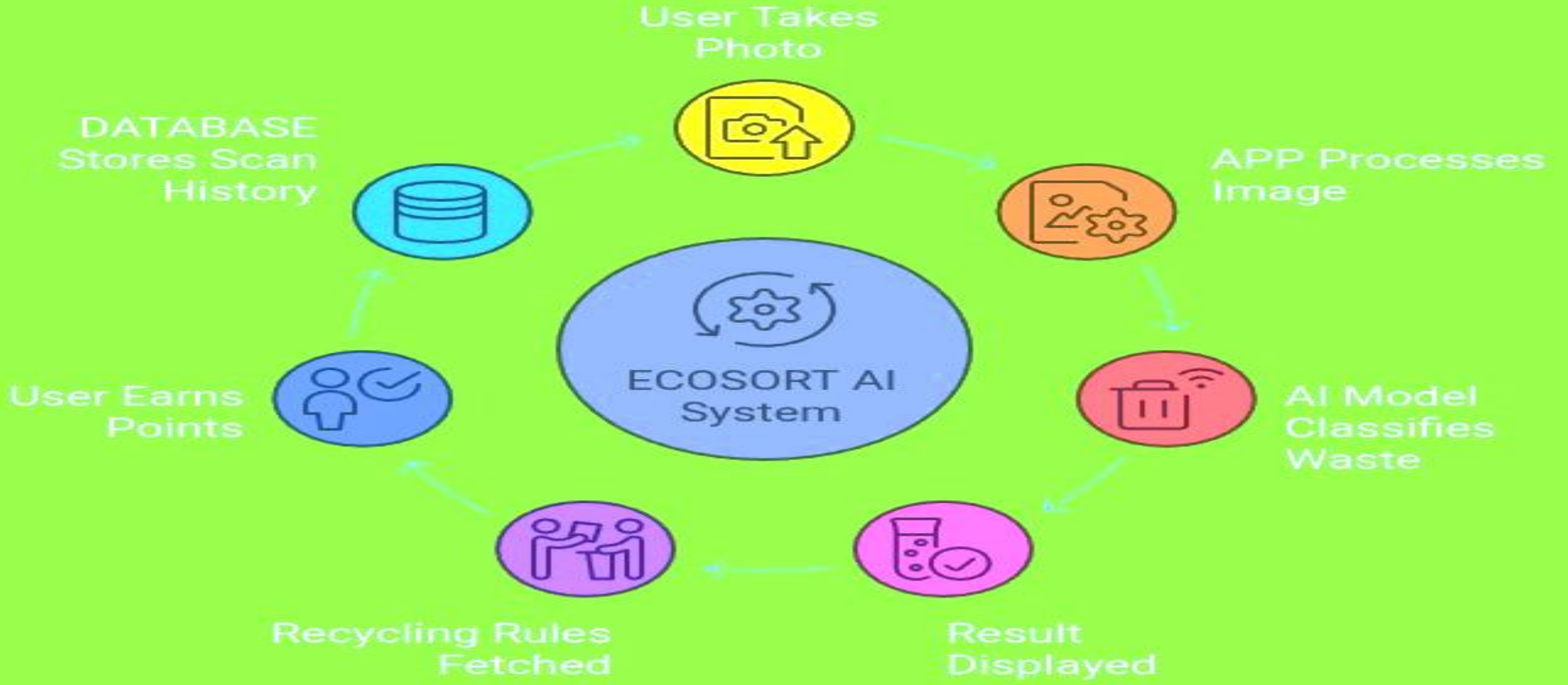
**What does the data contain?**
Raw images + labeled categories.

**Key variables/features:**

- Waste category
- Confidence score

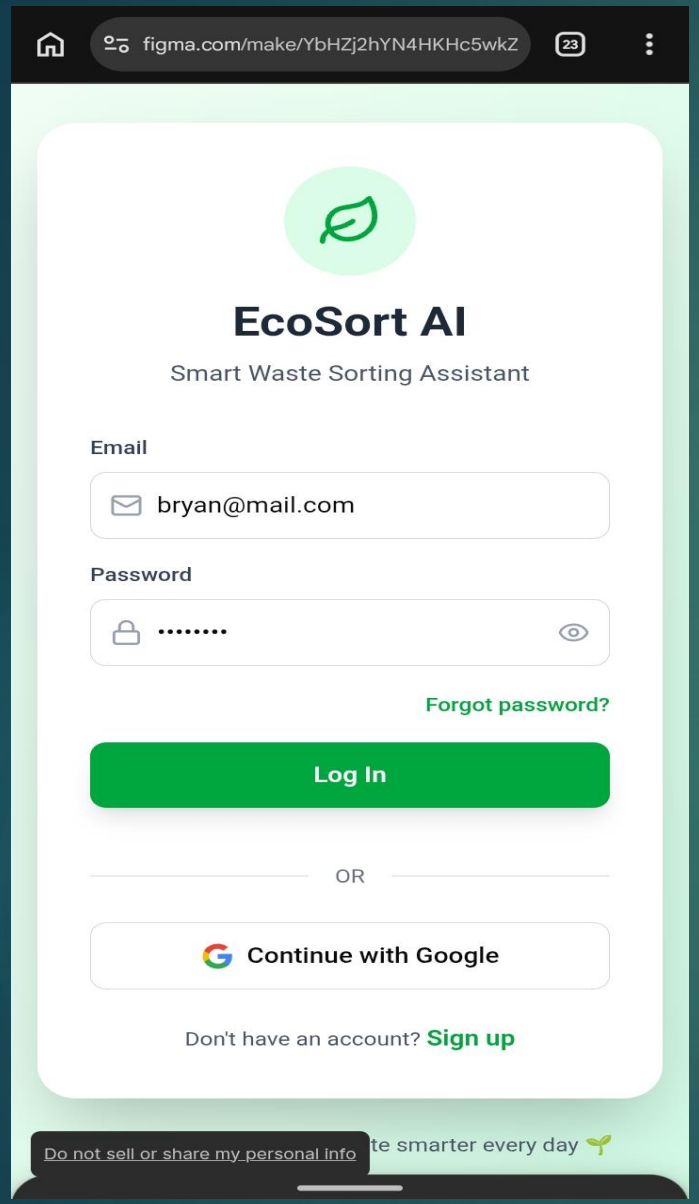
**Sample items recognized:**
Plastic bottles, glass bottles, metal ca
paper/cardboard, organic waste

# ECOSORT AI Waste Management Cycle




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# WIREFRAMES










https://www.figma.com/make/YbHZj2hYN4HKHc5wkZz0d9/ECOSORT-AI-App?t=OFShF950eZrA5piE-20&fullscreen=1

# SYSTEM COMPONENTS

## Mobile Application

A. Mobile Application

B.  Camera interface for capturing waste images

C. User registration and profile management

D.  Scan history display

E.  Points and rewards section

F.  Educational content in English and Kiswahili

## AI Engine

B. AI Engine

A. Tensor Flow Lite model running on-device

B. Recognizes 20+ Kenyan waste items

C. Confidence scoring

Continuous learning from user feedback

## C. Backend Server

A. User authentication

B. Database management

C. Recycling rules API for Kenyan counties

## DATABASE

Database

A. User information

B. Scan records

C. Recycling rules by location (county)

Achievement data

# **DESIGN THINKING**: Users

## Diverse Stakeholders in Waste Management




**Households & Individuals**

Everyday waste generators

**Businesses & Institutions**

Organizational waste producers

**Municipal Waste Collectors**

Waste collection services

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## Waste Management Challenges

What are the challenges faced by households and individuals regarding waste management?

Confusion about recyclables, lack of motivation, and limited access to disposal facilities.

What are the challenges faced by businesses and institutions?

High operational costs, difficulty training staff, and contaminated waste streams.

What are the challenges faced by municipal waste collectors?

Contaminated waste, limited resources, and difficulty engaging citizens.





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# CRITICAL THINKING TECHNIQUE

| TECHNIQUE | APPLICATION IN OUR PROJECT |
| --- | --- |
| Analysis | Studied Kenya's waste problem by category |
| Evaluation | Compared with existing apps(recycle coach) |
| Inference | Concluded AI can solve waste sorting confusion. |
| Problem-solving | Designed app with camera ,AI and sorting instructions |
| | |

# Assumption checking

Common assumptions in waste management:

- People will use the app consistently.
- AI can accurately identify all types of waste.
- Municipalities are ready to integrate digital solutions.
- Incentives alone will change behavior.

Critical check:

- Test whether users actually want real-time sorting guidance.
- Validate AI accuracy across diverse waste types (e.g., food packaging, electronics). el
- Confirm infrastructure readiness before scaling.
- Outcome: Avoid building features based on untested assumptions;

# Evidence-Based Thinking

Contamination rates in recycling streams can reach 25–30% when sorting is poor.

- Incentive-based programs (like deposit-return schemes) significantly improve compliance.

AI-powered image recognition has shown >90% accuracy in identifying recyclable materials in controlled environments.

Application to ECOSORT AI:

- Use evidence to design features (e.g., reward systems, AI image recognition). Continuously collect user data to refine algorithms and measure impact.

Benchmark against successful waste management models worldwide.

# ECOSORT AI WASTE APP Trade-offs

## ECOSORT AI WASTE APP Advantages

### Accurate Sorting

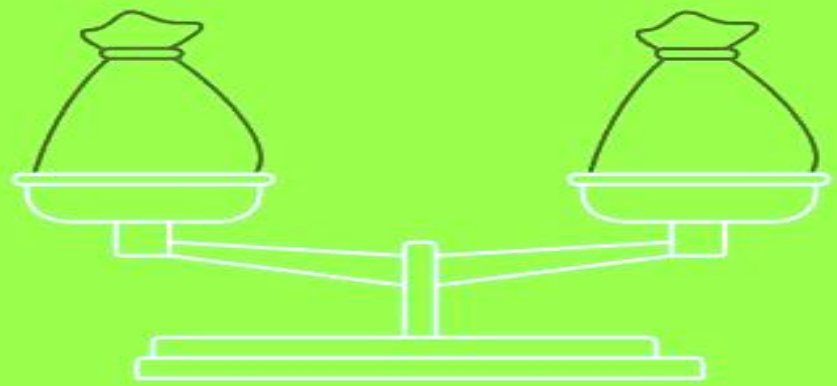
AI-powered recognition reduces contamination in recycling streams.

### User-Friendly Guidance

Provides real-time instructions for households and businesses, making waste disposal less confusing.

### Data Insights

Generates analytics on waste patterns, helping cities plan better collection routes and recycling strategies.



## ECOSORT AI WASTE APP Limitations

### Infrastructure Gaps

In areas with poor recycling facilities, even correct sorting may not lead to effective processing.

### Digital Divide

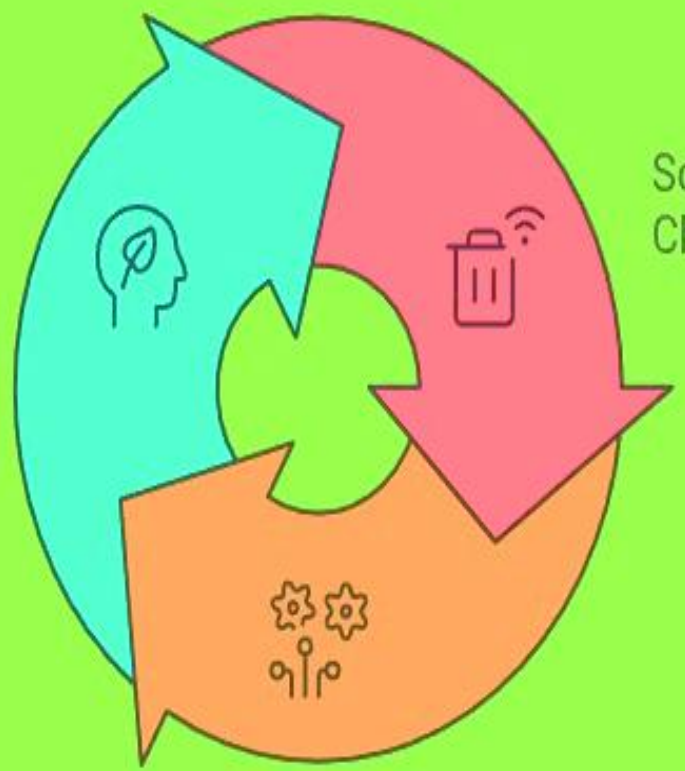
Requires smartphone access, which may exclude some individuals.

### Environmental Impact

Improves recycling rates, reduces landfill waste, and supports sustainability goals.

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# ECOSORT AI JUSTIFICATION



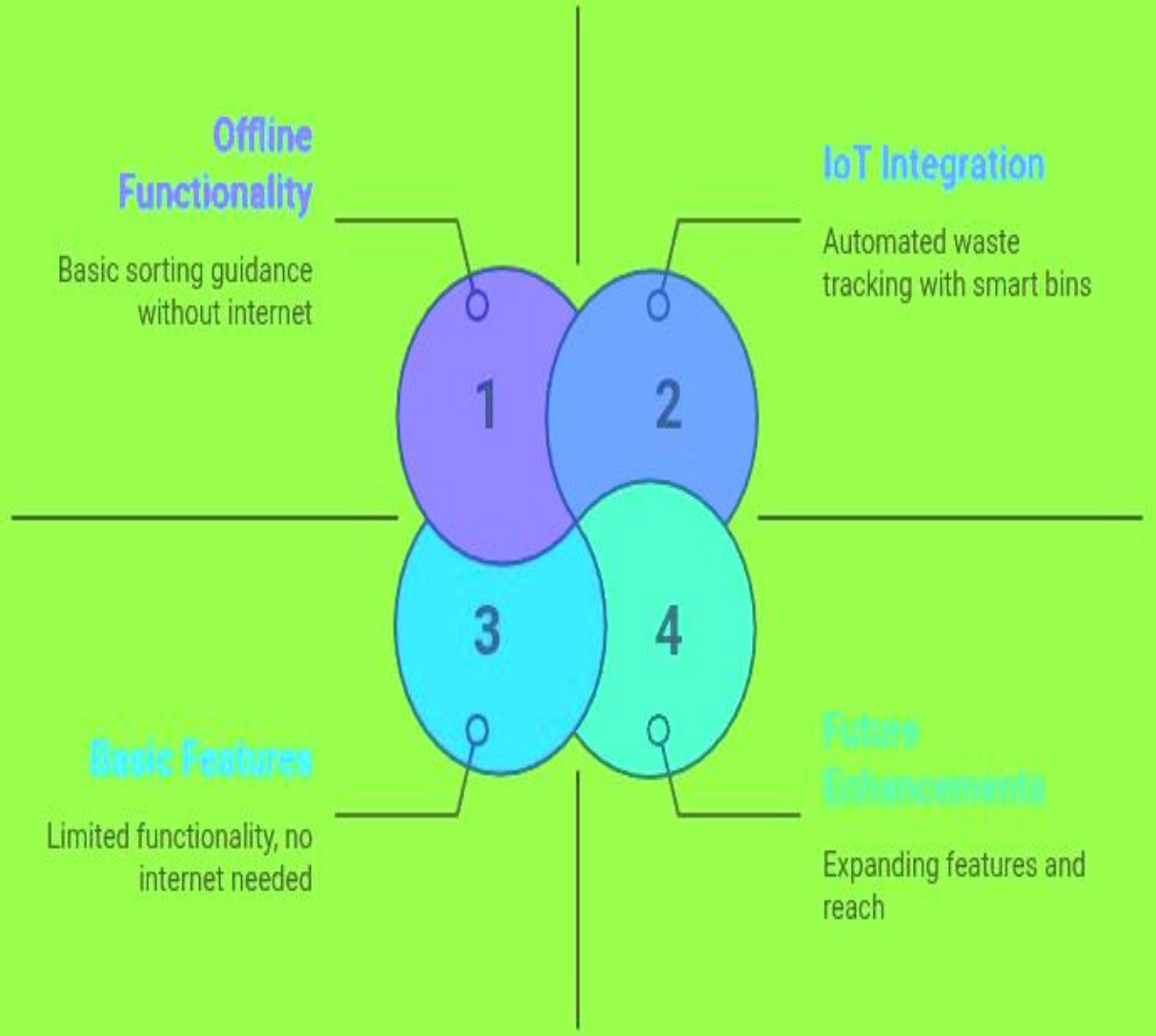
Promote Eco-Friendly Habits

Solve Waste Challenges

Leverage AI Technology

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# Improve Waste Management




**Offline Functionality**

Basic sorting guidance without internet

**IoT Integration**

Automated waste tracking with smart bins

**Basic Features**

Limited functionality, no internet needed

**Future Enhancements**

Expanding features and reach

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# Conclusion:
## ECOSORT AI Waste Management App

### Problem Summary

- Waste mismanagement is a pressing issue worldwide, driven by poor sorting practices, lack of awareness, contaminated recycling streams, and limited infrastructure. These challenges lead to environmental degradation, higher costs for municipalities, and inefficiencies in recycling systems.